Acoustics Projects
Review-I

Weekly Detailed Evaluation Sheets (Weeks 1 to 4)

Week 1: Orientation, Acoustics & Literature Breakdown

- **A. Problem Statement & Research Clarity (3 Marks):** Accurate extraction of abstract goals, inputs, outputs, and selected enhancement.
- **B. Literature Survey & Timelines (3 Marks):** Mapping 5–8 key prior works and establishing an 8-week SDLC milestone plan.
- **C. Environment & Audio Pipeline Setup (4 Marks):** Successful library installation (librosa, torchaudio), sample loading, waveform plotting, and resampling.
- **D. Feature Extraction & Exploration (5 Marks):** Computed STFT, Mel-spectrograms, CQT; generated pairwise similarity matrices and t-SNE embeddings.

Week 2: Dataset Collection, Splits & Preprocessing

- **A. Dataset Identification & Integrity (3 Marks):** Verified primary dataset source, licenses, file counts, and corruption filtering (dataset_stats.py).
- **B. Data Leakage Prevention & Splits (4 Marks):** Implemented author-specified train/val/test splits with fixed random seeds, ensuring zero speaker/artist overlap.
- **C. Preprocessing & Augmentation Pipeline (5 Marks):** Built complete feature extraction pipeline (normalization, .npy/.pt saving) and integrated paper-aligned augmentations (SpecAugment, pitch shift, noise).
- **D. DataLoader Optimization (3 Marks):** Verified PyTorch DataLoader throughput ($\geq$ 10 batches/sec) and tensor shape alignment.

Week 3: Architectural Analysis & Baseline Implementation

- **A. Architectural Breakdown (4 Marks):** Detailed data flow, component specification tables, and input/output tensor shapes matching the base paper.
- **B. Hyperparameters & Loss Formulation (4 Marks):** Documented loss equations, auxiliary losses, optimizers, learning rate schedulers, and mixed-precision (AMP) strategy.
- **C. Baseline Smoke Test & Verification (6 Marks):** Successfully trained a small 3-layer baseline (5 epochs) to verify loss descent, gradient flow, and parameter count via torchsummary.
- **D. First Full Training Run & Checkpointing (6 Marks):** Executed initial 10–20 epoch training run, logged learning curves, checked for NaN loss, and saved best model checkpoints.

Week 4: Training Experiments, Ablation & Test Evaluation

- **A. Hyperparameter Tuning & Grid Search (5 Marks):** Systematic exploration of learning rates, batch sizes, and loss weights with tracked logs.
- **B. Dataset & Augmentation Impact Analysis (4 Marks):** Evaluated performance differences across 50% vs. 100% data ratios and augmentation ON vs. OFF states.
- **C. Ablation Study Execution (5 Marks):** Re-trained top configuration through 3 structured ablation variants, measuring metric degradation against paper findings.
- **D. Test Results & Failure Case Analysis (6 Marks):** Evaluated top checkpoint on the held-out test set, compared metrics against paper benchmarks, and analyzed qualitative failure patterns / visualizations.

15-Minute Team Presentation Structure (Weeks 1 to 4)

Each team presentation during evaluation weeks must strictly adhere to the following timeline:

1. **Minutes 01–02:** Problem statement, paper breakdown, and selected project enhancement.
2. **Minutes 03–06:** Dataset verification, split validation, and preprocessing/augmentation pipeline walkthrough.
3. **Minutes 07–10:** Model architecture design, loss function formulation, and training/baseline execution logs.
4. **Minutes 11–13:** Hyperparameter tuning grids, ablation study results, and test evaluation metrics vs. paper benchmarks.

NOT GRADED

5. **Minutes 14–15:** Live code/repository walkthrough, failure case insights, and mentor Q&A defense.

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